

AI Career Pulse: Decoding the Job Market

Section 1: Project Initialization and Planning Phase

Aman Srivastava | SkillWallet Data Analytics Program

1. Project Initialization and Planning Phase

1.1 Background & Context

The artificial intelligence job market has grown at an unprecedented pace between 2025 and 2026, with new role types — Prompt Engineer, AI Agent Developer, Generative AI Engineer, Multimodal AI Engineer — emerging faster than standard compensation benchmarks can track them. Information about these roles is fragmented across job boards, salary-survey PDFs, and anecdotal social media posts, making it difficult for any single stakeholder to form an accurate, data-backed picture of the market.

This project, AI Career Pulse, was initiated to close that gap by building a single, interactive, filterable Business Intelligence tool in Tableau that turns raw AI job-posting data into decision-ready insight.

1.2 Problem Statement

There is no single, interactive, filterable view that lets a user answer practical questions such as:

- Which AI job category or title pays the most, on average and in total?
- Does company size (Startup vs. Big Tech) meaningfully change compensation?
- Which industries are hiring most aggressively for AI talent, beyond the obvious "Technology" sector?
- Does more experience actually translate into proportionally higher pay in AI roles?
- How is hiring volume trending year-over-year (2025 → 2026)?

Core problem statement: build and design a Tableau dashboard and story that consolidates AI job market data into clear, interactive, and filterable visualizations — enabling data-driven decisions for job seekers, recruiters, and analysts without requiring any manual spreadsheet analysis.

1.3 Project Objectives

1. Consolidate raw AI job-posting data into a single, clean Tableau data source.
2. Engineer calculated fields that convert raw numeric/text data into decision-ready metrics (e.g., salary tiers, demand scores, role-type flags).
3. Build a set of focused worksheets, each answering one specific business question.
4. Combine these worksheets into one interactive, filterable dashboard.
5. Construct a 5-page guided Story that walks a non-technical viewer through the same insights in a logical narrative order.
6. Extract and document clear, quantified insights that are useful to each of the three target stakeholders.

1.4 Target Stakeholders

Stakeholder	Key Question Answered
Job Seeker	Which AI role and skill set offers the best pay-to-entry-barrier ratio?
Recruiter / HR	What is a competitive salary band to offer for a given role, industry, and company size?

Industry Analyst	Where is AI hiring demand concentrated, and how is it shifting year over year?
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1.5 Scope & Constraints

- In scope: 91 AI job postings spanning January 2025 to 2026, covering 23 job titles across 11 job categories, 12 industries, 5 company-size bands, and 4 experience levels.
- Out of scope (for this iteration): non-AI job roles, salary data outside the captured Salary Tier bands, and real-time/live data refresh (the dataset is a static snapshot).
- Constraint: analysis relies on the accuracy and completeness of the source dataset; no external salary-survey data was blended in.

1.6 Tools & Environment

- Tableau Public / Tableau Desktop — for data connection, calculated fields, visualization, dashboarding, and storytelling.
- GitHub — for version control and hosting the project repository, including data files, workbook, documentation, and screenshots.

1.7 Repository Structure

Folder	Contents
data/	Raw and cleaned dataset(s) used for the analysis
workbook/	The Tableau (.twbx) workbook file
docs/	Project documentation, including this document set
assets/screenshots/	Exported screenshots of every worksheet, the dashboard, and the story

1.8 Project Phases & Planning Timeline

The project was planned and executed in five sequential phases, each building on the output of the previous one:

Phase	Key Activity	Output
1. Initialization & Planning	Define problem statement, objectives, stakeholders, and scope	This planning document
2. Data Collection & Preprocessing	Source data, clean fields, build calculated fields & filters	Cleaned, enriched Tableau data source
3. Data Visualization	Build 8 individual worksheets, one per business question	Sheet 3 – Sheet 10
4. Dashboard	Combine 5 core worksheets into one interactive dashboard	Dashboard 1
5. Story	Sequence key views into a 5-page guided narrative	Story 1